



AI reinvention made real

Accenture Innovation Challenge 2026

Team details

TEAM NAME: Warriors



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README: Implementation Approach & Methodology

- **1. Multi-Source Ingestion & Governed Canonical Contract:**

Harmonizes 3 mismatched enterprise grains (Daily ERP sales SQL, Hourly Web clickstream sessions, and Weekly Jira support tickets) into a unified daily canonical snapshot via MultiSourceDataLoader.

- **2. Deterministic Non-LLM Mathematical Core (Zero Hallucinations):**

- **SPC Deseasonalization:** 28-day rolling Day-of-Week (DoW) normalized baseline with 2.5-sigma threshold mathematically filters cyclical noise before alerting.

- **Exact Shapley Attribution:** Closed-form 3-factor metric tree decomposition ($\text{Revenue} = \text{Sessions} * \text{CVR} * \text{AOV}$) with guaranteed 0.00 residual error ($|\text{eps}| < 10^{-12}$) consuming exactly 0 LLM math tokens.

- **3. Triangulated AI Synthesis, Abstention & Security:**

- **3-Tier AI Loop:** Model 1 diagnoses on-premise Jira/ERP logs; Model 2 monitors live macro feeds; Model 3 performs attribution % and 30/60/90-day trajectory ROI simulation.

- **Explicit Abstention Protocol:** When competing cause margin is $< 25\%$, halts high-capital action and prescribes 2 low-cost canary tests (\$120-\$350).

- **Role-Based Access Control (RBAC):** Masks sensitive unit COGS and gross margins for Operations Analysts while granting full access to Executives.

README: Solution Architecture & Engine Layers

Layer 1: Governed Ingestion & Schema Contracts (engine/contracts/, engine/data/)

- **Canonical Data Harmonization:** Ingests daily ERP sales, hourly web sessions, and weekly Jira tickets into validated Pydantic schemas.
- **Dynamic RBAC Masking Engine:** Performs real-time column/row redaction on financial margins based on active user persona token.

Layer 2: Deterministic Non-LLM Math Core (engine/math/)

- **Statistical Process Control (spc.py):** 28-day DoW baseline, Student-t cold-start envelope ($N < 14$), and MAD robust outlier detection.
- **Axiomatic Causal Metric Tree (causal_tree.py):** Exact 3-factor Shapley attribution (Sessions, CVR, AOV) + LMDI-1 validation.

Layer 3: 3-Model AI Synthesis & Simulation (engine/synthesis/)

- **Model 1 (Internal Diagnostic):** Private on-premise LLM diagnosing warehouse backlogs and Jira incident tickets.
- **Model 2 (Macro Sentinel):** Live API sentinel ingesting port strikes, freight indices, and competitor campaigns.
- **Model 3 (Prescriptive Trajectory Simulator):** Attribution weighting + 30/60/90-day ROI curves under executive budget caps (\$45k).
- **Abstention Engine (abstention.py):** Calculates hypothesis confidence delta; triggers canary validation tests if margin $< 25\%$.

Layer 4: Interactive Decision Workspace & Telemetry (ui/, engine/telemetry/)

- **Streamlit UI & Telemetry Tracker:** 6-tab workspace with Plotly charts, persona switchers, human feedback loop, and ms latency/token accounting.

README: Dependencies & Technology Ecosystem

Runtime Core & Execution Environment:

- **Python 3.11+ / PyPy**: Modern Python runtime utilizing typing, dataclasses, and async execution.
- **Pydantic v2 (>=2.5.0)**: Strict runtime schema enforcement, data contract validation, and JSON serialization.

Front-End Interactive Workspace & Visualization:

- **Streamlit (>=1.30.0)**: Interactive multi-tab executive dashboard framework with session state management.
- **Plotly (>=5.18.0)**: Interactive WebGL/SVG visualization engine powering SPC Control Charts, Shapley Waterfall, and 30/60/90-Day Trajectory ROI curves.

Data Processing & Mathematical Computation:

- **Pandas (>=2.1.0)**: High-performance time-series alignment, daily grain resampling, and DataFrame RBAC masking.
- **NumPy (>=1.26.0)**: Vectorized numerical routines, Student-t distribution limits, and Median Absolute Deviation (MAD).

Pluggable Multi-LLM Provider Architecture:

- **Hybrid Provider Interface**: Pluggable provider adapter supporting Google Gemini, OpenAI GPT-4o, and local Ollama.
- **Deterministic Offline Mock Fallback**: Built-in deterministic mock provider for instant execution with <5ms latency and \$0.00 token cost.

Verification & Testing Infrastructure:

- **Python unittest & Monte Carlo**: Headless test harness executing 142 assertions and 10,000 randomized invariant stress trials.

1. Environment Setup & Dependency Installation:

- **Terminal Command:** `pip install -r prototype/requirements.txt`
- **Prerequisites:** Python 3.11+ installed; virtual environment recommended.

2. Launch Interactive Decision Workspace UI:

- **Terminal Command:** `streamlit run prototype/app.py`
- **Browser Access:** Open `http://localhost:8501` (Interactive scenarios, persona toggles, trajectory sliders).

3. Run Automated Headless Verification Test Suite:

- **Acceptance Tests (93/93 Passed):** `python prototype/test_scenarios.py`
- **Modular Unit & Property Tests (49/49 Passed):** `python -m unittest discover -s prototype/tests -p "test_*.py"`
- **Scenario x Persona Matrix Runner (8/8 Passed):** `python prototype/verify_e2e.py`

4. Key Verification Guarantees & Acceptance Matrix:

- **Deterministic Math Invariant:** Shapley metric tree residual error 0.00000000 (jews) $< 10^{-12}$ across 10,000 Monte Carlo trials.
- **Zero Math Token Invariant:** Statistical Process Control and Causal Tree attribution consume exactly 0 LLM tokens (\$0.00 cost).
- **All 4 Mandatory Scenarios:** 1. Multifactor (70/70) | 2. Ambiguity & Canary tests | 3. Cold start prior | 4. RBAC masking.